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HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma 2025 Session

Specialty/option: SFTWARE ENGINEERING (SWE)
Paper: Information systems
Duration: 3 hours

Credits: 6

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*
- *Where calculations are required, clearly show your working and be chronological in your answer.*

SECTION A: MULTIPLE CHOICE QUESTIONS (40 MARKS)

One (01) Mark for Each Correct Answer. Write Your Answers in Your Booklet

- 1) Which of the following is not a characteristic of good test data
 - a) users do not participate at this preliminary stage
 - b) should be comprehensive
 - c) every statement should be executed
 - d) All of the above
- 2) In the system concepts, term Integration
 - a) implies structure and order
 - b) refers to the manner in which each component functions with other components of the system.
 - c) means that parts of the computer system depend on one another.
 - d) refers to the holism of system
- 3) The rule(s) to follow in constructing decision tables is (are):
 - a) a decision should be given a name
 - b) the logic of the table is independent of the sequence in which conditions rules are written, but the action takes place in the order in which the events occur.
 - c) standardized language must be used consistently.
 - d) All of the above
- 4) is a group of interested components working together towards a common goal by accepting inputs and producing outputs in an organized transformation process.
 - a) System
 - b) Network
 - c) Team
 - d) System Unit

- 5) A rectangle in a DFD represents
- a process
 - a data store
 - an external entity
 - an input unit
- 6) External Entities may be a
- Source of input data only
 - Source of input data or destination of results
 - Destination of results only
 - Repository of data
- 7) The major goal of requirement determination phase of information system development is
- determine whether information is needed by an organization
 - determine what information is needed by an organization
 - determine how information needed by an organization can be provided
 - determine when information is to be given
- 8) It is necessary to prioritize information requirements of an organization at the requirements determination phase as
- it is always good to prioritize
 - there are conflicting demands from users
 - there are constraints on budgets, available time, human resource and requirement
 - all good organization do it
- 9) Requirement specification is carried out
- after requirements are determined
 - before requirements are determined
 - simultaneously with requirements determination
 - independent of requirements determination
- 10) The role of a system analyst drawing up a requirements specification is similar to
- architect designing a building
 - a structural engineer designing a building
 - a contractor constructing a building
 - the workers who construct a building
- 11) It is necessary to consult the following while drawing up requirement specification
- only top managers
 - only top and middle management
 - only top, middle and operational managers
 - top, middle and operational managers and also all who will use the system
- 12) In order to understand the working of an organization for which a computer based system is being designed, an analyst must
- look at only current work and document flow in the organization
 - discuss with top level and middle level management only
 - interview top, middle, line managers and also clerks who will enter data and use the system
 -

- e) only clerical and middle level staff who have long experience in the organization and will be users of the system
- 13) A feasibility study is carried out**
 - a) after final requirements specifications are drawn up
 - b) during the period when requirements specifications are drawn up
 - c) before the final requirements specifications are drawn up
 - d) at any time
- 14) The main objective of feasibility study is**
 - a) to assess whether it is possible to meet the requirements specifications
 - b) to assess if it is possible to meet the requirements specified subject to constraints of budget, human resource and hardware
 - c) to assist the management in implementing the desired system
 - d) to remove bottlenecks in implementing the desired system
- 15) It is necessary to carry out a feasibility study as**
 - a) top management cannot ensure that a project is feasible before calling a system analyst
 - b) top management is not sure what they want from the system
 - c) even though top management is in favour of the system, technology may not be mature for implementation
 - d) all organizations do it
- 16) Feasibility study is carried out by**
 - a) managers of the organization
 - b) system analyst in consultation with managers of the organization
 - c) users of the proposed system
 - d) systems designers in consultation with the prospective users of the system
- 17) Initial requirements specification is**
 - a) not changed till the end of the project
 - b) continuously changed during project implementation
 - c) only a rough indication of the requirement
 - d) changed and finalized after feasibility study
- 18) Final specifications are drawn up by**
 - a) system analyst in consultation with the management of the organization
 - b) the managers of user organization
 - c) system analyst in consultation with programmers
 - d) system designers along with users
- 19) The main goal of arriving at a final specification is**
 - a) to tell the organization's managers how the system will function.
 - b) to tell the organization's managers what the proposed system will achieve in a language understood by them.
 - c) to compute the cost of implementing the system.
 - d) to assist in designing the system.
- 20) The final specifications are arrived at**
 - a) after feasibility study
 - b) during feasibility study

- c) just before implementation phase
 - d) when the system is being designed
- 21) System approval criteria are specified**
- a) when the final specifications are drawn up
 - b) during feasibility study
 - c) during the requirements specifications stage
 - d) during system study stage
- 22) Hardware study is required**
- a) to find out cost of computer system needed
 - b) to determine the type of computer system and software tools needed to meet the final system specification
 - c) to make sure that the system does not become obsolete
 - d) to find how to implement the system
- 23) Hardware study is carried out**
- a) after the final system is specified
 - b) at the requirements specification stage
 - c) before the requirements are specified
 - d) whenever management decides it is necessary
- 24) System design is carried out**
- a) as soon as system requirements are determined
 - b) whenever a system analyst feels it is urgent
 - c) after final system specifications are approved by the organization
 - d) whenever the user management feels it should be done
- 25) The primary objective of system design is to**
- a) design the programs, databases and test plan
 - b) design only user interfaces
 - c) implement the system
 - d) find out how the system will perform
- 26) The primary objective of system implementation is**
- i) to build a system prototype
 - ii) to train users to operate the system
 - iii) to implement designed system using computers
 - iv) write programs, create databases and test with live data
- a) i, iii
 - b) i, ii, iii
 - c) ii, iii
 - d) ii, iv
- 27) During system implementation the following are done**
- i) programs are written and tested with operational data
 - ii) user documentation is created and users trained
 - iii) programmers are recruited and trained
 - iv) the system is tested with operational data
- a) i and iii
 - b) ii and iii

- c) ii and iv
 - d) i, ii & iv
- 28) System evaluation is carried out**
- a) after the system has been operational for a reasonable time
 - b) during system implementation
 - c) whenever managers of user organization want it
 - d) whenever operational staff want it
- 29) The main objective of system evaluation is**
- a) to see whether the system met specification
 - b) to improve the system based on operational experience for a period
 - c) to remove bugs in the programs
 - d) to assess the efficiency of the system
- 30) Systems are modified whenever**
- a) user's requirements change
 - b) new computers are introduced in the market
 - c) new software tools become available in the market
 - d) other similar organization modify these system
- 31) The main objective of system modification is**
- a) to use the latest software tools
 - b) to meet the user's new/changed needs
 - c) to use the latest hardware
 - d) to have the most modern system
- 32) To easily modify the existing system it is necessary to**
- a) use good software tools
 - b) use the best hardware available
 - c) design the system which can be changed at low cost
 - d) keep the programming team happy
- 33) It is necessary to design an information system to easily accommodate change, because**
- a) new computers are introduced every year
 - b) new computer languages become popular every year
 - c) organizations' requirements change over a period of time
 - d) systems need continuous debugging
- 34) Changing an operational information system is**
- a) impossible
 - b) expensive and done selectively
 - c) never required
 - d) usually done
- 35) System analysts have to interact with**
- i) managers of organizations
 - ii) users in the organization ✓
 - iii) programming team ✓
 - iv) data entry operator ✓

- a) iii and iv
 - b) i, ii and iii
 - c) ii, iii and iv
 - d) ii and iii
- 36) The primary responsibility of a systems analyst is to**
- a) specify an information system which meets the requirements of an organization
 - b) write programs to meet specifications
 - c) maintain the system
 - d) meet managers of the organization regularly
- 37) The responsibilities of a system analyst include**
- i) defining and prioritizing information requirement of an organization ✓
 - ii) gathering data, facts and opinions of users in an organization ✓
 - iii) drawing up specifications of the system for an organization ✓
 - iv) designing and evaluating the system
- a) i and ii
 - b) i, ii and iv
 - c) i, ii, iii and iv
 - d) i, ii and iii
- 38) The most important attribute of a systems analyst is**
- a) excellent programming skills
 - b) very good hardware designing skills
 - c) very good technical management skills
 - d) very good writing skills
- 39) Among the attributes of a good systems analyst the following are essential**
- i) knowledge of organization ✓
 - ii) analytical mind ✓
 - iii) ability to communicate orally ✓
 - iv) excellent mathematical abilities
- a) i and ii
 - b) i, ii and iii
 - c) i, ii and iv
 - d) i, iii and iv
- 40) Among the attributes of a systems analyst the following are most important :**
- i) knowledge of computer systems and currently available hardware ✓
 - ii) good interpersonal relations
 - iii) broad knowledge about various organizations
 - iv) very good accountancy knowledge
- a) i, iii and iv
 - b) i and iii
 - c) i, ii and iv
 - d) i, ii and iii

SECTION B: OPERATING SYSTEMS.

(40 MARKS)

PART I: MCQS

20 Marks

- 1) Dual mode of operating system has
 - a) 1 mode
 - ~~b) 2 modes~~
 - c) 3 modes
 - d) 4 modes
- 2) Multi-processor system gives a
 - a) small system
 - b) tightly coupled system
 - c) loosely coupled system
 - d) both a and b
- 3) Logical extension of multiprogramming operating system is
 - a) time sharing
 - b) multi-tasking
 - c) single programing
 - d) both a and b
- 4) Multiprocessor system have advantage of
 - a) Increased Throughput
 - b) Expensive hardware
 - c) operating system
 - d) both a and b
- 5) Scheduling of threads are done by
 - a) input
 - b) output
 - c) operating system
 - d) memory
- 6) Multiprogramming of computer system increases
 - a) memory
 - b) storage
 - c) CPU utilization
 - d) cost
- 7) Another type of multiple-CPU system is the
 - a) mini Computer
 - b) Super Computer
 - c) Clustered System
 - d) both a and b
- 8) Interrupt table of pointers having addresses for each interrupt is located at
 - a) high memory
 - b) low memory
 - c) mid memory
 - d) both a and b
- 9) Example of open source operating system is
 - a) UNIX
 - b) Linux
 - c) windows
 - ~~d) both a and b~~
- 10) Main memory of computer system is also called
 - ~~a) non volatile~~
 - b) volatile
 - c) reserved

- d) large
- 11) Controller of computer system transfers data from device to
- buffers
 - cache
 - registers
 - indexes
- 12) When many users accesses mainframes, this approach is called as
- resource allocation
 - word processors
 - dedicated resources
 - interface
- 13) Accessing same data from storage of computer system is provided by
- serial clusters
 - parallel clusters
 - Beowulf clusters
 - ~~a)~~ both a and b
- 14) To start an I/O operation device driver loads appropriate register into?
- memory
 - Secondary storage
 - Device Controller
 - Arrays
- 15) Symmetric multiprocessing architecture of computer system uses shared
- bus
 - memory
 - processors
 - ~~a)~~ both a and b
- 16) In asymmetric clustering other machines perform operations while one machine is in
- hot standby mode
 - standby mode
 - reset mode
 - undefined mode
- 17) Ability to continuously providing service proportional to level of surviving hardware
- graceful upgradation
 - degradation
 - upgradation
 - graceful degradation
 - e)
- 18) Secondary memory of computer system is also called
- non volatile
 - ~~a)~~ volatile
 - reserved
 - small
- 19) Environment in which programs of computer system are executed is:
- ~~a)~~ operating system
 - nodes
 - clustered system
 - both a and b
- 20) One word memory storage is collection of
- 2 bytes
 - 4 bytes
 - 7 bytes
 - 8 bytes

Part II Structural Questions

20 Marks

1. List the four (04) key components of android architecture. 2 marks
2. Differentiate between general purpose operating system and mobile operating system. 3 marks
3. Describe the operating system operations. 3 marks
4. Explain simple batch system. 2 marks
5. Explain time sharing operating system. 2 marks
6. List out any four process control system calls. 3 marks
7. Explain virtual machines. 2 marks
8. What is the Google Android SDK? 3 marks

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 MARKS)

Part I: MCQs

10 Marks

- 1) Which of these is not one of the constraints of a project?
 - a) Scope
 - b) Resources
 - ~~c) Team~~
 - d) Budget
- 2) Which of the following is not correct about initial phase of a project?
 - a) The cost associated at the beginning of the project is highest.
 - b) Stakeholders have maximum influence during this phase
 - c) The highest uncertainty is at this stage of the project.
 - d) All the above statements are correct.
- 3) The project you are managing has nine stakeholders. How many channel of communications are there between these stakeholders?
 - a) 9
 - b) 8
 - c) 45
 - d) 36
- 4) Which of the following is not an example of formal communication?
 - a) Contract
 - b) email
 - c) Project status report
 - d) Status meeting
- 5) A Project with a total funding of \$100,000 finished with a BAC value of \$95,000. What term can BEST describe the difference of \$5,000?
 - a) Cost Variance
 - b) Management Overhead
 - ~~c) Management Contingency Reserve~~
 - d) Schedule Variance
- 6) If the Earned Value is equal to Actual Cost, it means:
 - a) Project is on budget and on schedule
 - b) Schedule Variance Index is 1
 - c) There is no schedule variance
 - d) There is no cost variance
- 7) Which of the following is the most important element of Project Management Plan that is useful in HR Planning process:
 - a) Risk Management activities

- b) Quality Assurance activities
 - c) Activity Resource requirements
 - d) Budget Control activities
- 8) Which of the following types of Organizational Charts can be BEST used to track project costs :
- a) Hierarchical-type Organizational Chart
 - b) Organizational Breakdown Structure
 - c) Resource Breakdown Structure
 - d) Responsibility Assignment Matrix
- 9) Which of the following is not true for a Responsibility Assignment Matrix (RAM):
- a) Defines what project team is responsible for each WBS component
 - b) Defines what project member is responsible for each activity
 - c) Text description of roles, responsibilities, authorities
 - d) Defines all people associated with each activity
- 10) Process Analysis is a function of:
- a) Performance Analysis
 - b) Quality Metrics
 - c) Process Improvement Plan
 - d) Quality Improvement Plan

Part II Structural Questions

10 Marks

- 1) Software project management comprises of a number of activities, which contains planning of project, deciding scope of software product, estimation of cost in various terms, scheduling of tasks and events, and resource management.
 - a) What are the activities that are carried out during project scope management activities **3marks**
 - b) Explain what is usually estimated during project estimation phase. **2 marks**
 - c) Discuss two types of Project Estimation Techniques. **1 mark**
- 2) Differentiate between agile project management and prince2. **2 marks**
- 3) Define software law. **2 marks**

a) - delivery